

Memory Systems Architecture
Assignment #1
Due: Tuesday, Jan 20, 11:59 pm
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Introduction: This is an individual assignment. It is meant to test your knowledge of caching and your ability to do experimental evaluations. You can use the starter file `problem1.cpp` to get started.

Problem:

Write a C/C++ program to determine the (a) number of levels in the cache hierarchy (b) linesize (c) capacity of each level of cache, and (d) the (approximate) memory latency of the computer system.

- i. Report the results for one machine (say your own laptop)
- ii. Write a report explaining the kernels you used for each of the three parts

A submission that reports the correct numbers without explaining the insights used in obtaining them will not get significant credit.

While writing the code you must be careful about microarchitectural enhancements such as out-of-order execution and prefetching that may affect memory system behavior.

Submission Files:

A single file named **problem1.cpp** that contains code for all three parts.

A report file named **report1.pdf** (maximum two pages)

Extra Credit:

Write a kernel to determine the associativity of the last level cache.